

Table 3: Answer table for Q.4 over Table 3 of Piechulla et al. [51], modeled in the ORKG as Comparison R1469383.

contrib.	phosphor	coating	precursors	temp (°C)	thick. (nm)
R1469207	$\text{Sr}_2\text{Si}_5\text{N}_8\text{:Eu}^{2+}$	Al_2O_3	TMA/ H_2O	30	8
R1469207	$\text{Sr}_2\text{Si}_5\text{N}_8\text{:Eu}^{2+}$	Al_2O_3	TMA/ O_3	30	8

Complex Query. The fourth question is: “Among Eu^{2+} -doped red phosphors, which ALD coatings deposited at $\leq 150^\circ\text{C}$ achieved an optimal thickness of $\leq 20\text{ nm}$, and which precursors were used?” The SPARQL query implementation for this natural language question can be rerun via <https://tinyurl.com/Red-Eu2-ALD-thinlowT>. As explained earlier, answers in this work are tabulated, synthesized results based on the original question; Table 3 lists all Eu^{2+} -doped red phosphors, coatings, precursors, deposition temperatures, and optimal thicknesses that satisfy these query conditions.

3.2.2 Machine-actionable modeling of Table 4

The next ORKG comparison, available at <https://orkg.org/comparisons/R1469594> [49], is modeled after Table 4 of Piechulla et al. (2025) [51]. This table compiles reported thickness ranges and growth-per-cycle (GPC) values for ALD coatings on pharmaceutically relevant particulate supports. It contains 25 experimental entries from ten publications, covering a variety of drug carriers and biomolecular substrates (e.g., lactose, budesonide, indomethacin, HPV capsomers, and proteins), together with ALD process parameters such as precursor chemistries, deposition temperatures, coating thicknesses, and GPC values. In the ORKG, this information is represented as a machine-actionable comparison that underpins the question-answering examples in this section.

Table 4: Answer table for Q.5 over Table 4 of Piechulla et al. [51], modeled in the ORKG as Comparison R1469594. Rows with identical support, precursor pair, and temperature are merged for compactness.

support	precursors	temp (°C)	thick. (nm)	contrib.
budesonide	$\text{SiCl}_4/\text{H}_2\text{O}$	40	1.5–10.0	R1469559, R1469560, R1469558
budesonide	TMA/ H_2O	40	0.8–2.0	R1469545
budesonide	TMA/ O_3	40	20.0–50.0	R1469551, R1469548, R1469554
budesonide	$\text{TiCl}_4/\text{H}_2\text{O}$	40	3.0–15.0	R1469557, R1469555, R1469556
lactose	TMA/ H_2O	30	2.5–11.0	R1469543, R1469540
lactose	TMA/ O_3	30	3.8–10.3	R1469544
myoglobin	TMA/ H_2O	30	8.3	R1469580
posaconazole/ HPMCAS	TMA/ H_2O	35	8.0–40.0	R1469589
posaconazole/ HPMCAS	TMA/ H_2O	35	12	R1469588

Easy Query. The fifth question is: “Which support materials were coated at $\leq 40^\circ\text{C}$, and which precursor pairs were used, along with the reported coating thickness?” The SPARQL query implementation for this question can be rerun via <https://tinyurl.com/less-than-40>. The tabulated answer is in Table 4, which lists all pharmaceutically relevant supports, precursor pairs, deposition temperatures, and thickness ranges that satisfy the $\leq 40^\circ\text{C}$ constraint.

Table 5: Answer table for Q.6 over Table 4 of Piechulla et al. [51], modeled in the ORKG as Comparison R1469594. Rows with identical support–precursor–temperature combinations are merged for compactness.

support	precursors	temp (°C)	thick. (nm)	GPC (nm)	class	contrib.
lactose	TMA/H ₂ O	30	2.5	0.47	average	R1469540, R1469543
lactose	TMA/O ₃	30	3.8	0.47	average	R1469544
budesonide	TMA/H ₂ O	40	0.8	0.20	slow	R1469545
budesonide	TiCl ₄ /H ₂ O	40	3.0	0.30	n/a	R1469555, R1469556, R1469557
budesonide	SiCl ₄ /H ₂ O	40	1.5	0.10	n/a	R1469558, R1469559, R1469560
myoglobin	TMA/H ₂ O	30	8.3	0.21	slow	R1469580
posaconazole/TMA/H ₂ O		35	12	0.40	average	R1469588
HPMCAS						
posaconazole/TMA/H ₂ O		35	8.0	0.40	average	R1469589
HPMCAS						

Complex Query. The sixth question is: “Among low-temperature runs (< 70 °C) that produced thin coatings (< 20 nm), which support materials, precursor pairs, temperatures, thicknesses, and growth-per-cycle (GPC) values were reported, and how do the GPC values classify as slow, average, or fast for alumina coatings?” The SPARQL query implementation can be rerun via <https://tinyurl.com/pharma-hard-query>. The tabulated answer is in Table 5, which lists all support materials, precursor pairs, temperatures, thicknesses, GPC values, and corresponding GPC classes satisfying the query constraints.

3.3 Paper – Atomic and molecular layer deposition of functional thin films based on rare earth elements

We now consider the review by Ghazy *et al.* (2025) [25], titled “Atomic and molecular layer deposition of functional thin films based on rare earth elements.” In contrast to the first paper, which focused on conformality in high-aspect-ratio test structures, and the second, which provided a large-scale quantitative synthesis of particulate ALD systems, this review centers on the chemistry and functional properties of rare-earth-based ALD and MLD thin films. From this paper, we model Tables 2, 3, 4, and 5 as machine-actionable ORKG comparisons, each accompanied by Easy and Complex Queries. Together with the previous two review papers, these models cover a broad spectrum of ALD-related information and show that diverse experimental and materials data can be represented within a unified, queryable knowledge graph framework in the ORKG.

3.3.1 Machine-actionable modeling of Table 2

We first convert Table 2 of Ghazy *et al.* (2025) into an ORKG comparison, available at <https://orkg.org/comparisons/R1469955> [1]. This makes it citable, reusable, and versionable as a *living review table* with full provenance and access to earlier versions. Table 2, titled “*Use of different R species as dopants in ALD*”, lists host materials and rare-earth dopants used in ALD processes. It spans oxides (e.g., Al₂O₃, TiO₂, HfO₂), sulfides (e.g., ZnS, SrS, CaS), and garnets (e.g., Y₃Ga₅O₁₂, Lu₃Al₅O₁₂), together with reported applications such as luminescence, magnetics, gate dielectrics, electrolytes, and thermometry. Each record links a specific host–dopant combination to one or more literature references, providing the structured basis for the Easy and Complex Queries discussed next.

Easy Query. The seventh question is: “For each application (luminescence, memory, gate dielectrics, electrolytes, etc.), which rare-earth dopants appear on the largest number of distinct hosts?” The SPARQL query implementation for this question can be rerun via <https://tinyurl.com/versatile-dopants> over the ORKG comparison R1469955 of Table 2 in the review article. The tabulated answer is in Table 6, which lists, for each application, the rare-earth dopants and their corresponding host counts.

Table 6: Answer table for Q.7 over Table 2 of Ghazy et al. [25], modeled in the ORKG as Comparison R1469955. For each application, the table lists rare-earth dopants and the number of distinct host materials.

application	dopant(s)	hosts
Electrolyte	Y, Dy, La	2, 1, 1
Gate dielectrics	La, Dy, Er, Gd, Pr, Y	2, 1, 1, 1, 1, 1
Interphase	Dy, La, Y	1, 1, 1
Luminescence	Er, Eu, Tb, Nd, Ce, Dy, Ho, La, Pr, Sm, Tm, Yb, (Er,Yb)	12, 5, 5, 3, 2, 2, 2, 2, 2, 2, 2, 1
Magnetics	Dy, Er, Eu, Ho, La, Nd, Pr, Sm, Tb, Tm, Yb	1 each
Memory	Gd, La, Y, Dy, Er, Pr	2, 2, 2, 1, 1, 1
Optoelectronics	La	1
Thermometry	Dy, Er, Eu, Ho, Nd, Sm, Tb, Tm, Yb	1 each
Waveguides	(Er,Yb), Er, Yb	1 each

Table 7: Answer table for Q.8 over Table 2 of Ghazy et al. [25], modeled in the ORKG as Comparison R1469955. For each host material appearing in at least two application domains, the table lists its applications and rare-earth dopants.

host	applications	dopants
Al ₂ O ₃	Luminescence, Waveguides	(Er,Yb), Er, Yb
HfO ₂	Gate dielectrics, Memory	La, Dy, Er, Gd, Pr, Y
TiO ₂	Luminescence, Magnetics	Er, Eu, Dy, Ho, La, Nd, Pr, Sm, Tb, Tm, Yb
ZrO ₂	Electrolyte, Interphase	Dy, La, Y

Complex Query. The eighth question is: “Which host materials appear in two or more different application domains, and with which dopants?” The SPARQL query implementation for this question can be rerun via <https://tinyurl.com/cross-functional-hosts> over the ORKG comparison R1469955 of Table 2 in the review article. The tabulated answer is in Table 7, which lists all host materials occurring in at least two application domains, together with their applications and associated rare-earth dopants.

3.3.2 Machine-actionable modeling of Table 3

We modeled Table 3 of Ghazy *et al.* (2025) as an ORKG comparison available at <https://orkg.org/comparisons/R1471077> [2]. The comparison records, for each reported rare-earth (R³⁺) ALD process, the deposited material, metal precursor(s), co-reactant(s), growth per cycle (GPC), deposition temperature window, and literature references, yielding a structured representation of 254 compared ALD processes. The table spans a broad range of binary and ternary systems, including rare-earth combinations with Al, Ti, Ni, Ga, W, Co, Mn, Fe, Hf, Y, Sc, Lu, and Li, many of which employ supercycle strategies to control element ratios. In the ORKG, this information becomes a machine-actionable basis for the Easy and Complex Queries on rare-earth ALD chemistry introduced next.

Easy Query. The ninth question is: “List all rare-earth ALD processes that achieve high growth per cycle (GPC ≥ 1 Å) at low deposition temperature (≤ 250 °C). Report material, metal precursor family, co-reactant, GPC, and temperature; sort by GPC.” The SPARQL query implementation for this question can be rerun via <https://tinyurl.com/alld-highgpc-lowtemp> over the ORKG comparison R1471077 of Table 3 in the review article. The tabulated answer is in Table 8, which